ML

- AI vs ML
- AI = ML
- AI <> ML

X

ML

involves predictions or decisions based on data.

- supervised machine learning.
- unsupervised machine learning.

AI
↓
ML
↓
NN
↓
DL

Factor influence decision of a model

1) inputs
2) assign weights
3) calculations (input * assign weights)

Positive → Take The decision

example

Imagine you are trying to decide whether to order a pizza for dinner. You might create a mental checklist of questions:
Will it save time? (Yes)
Will I lose weight? (No)
Will it save money? (Yes, because I have a coupon)

You then decide how important each question is to you (giving them "weights").

Machine Learning usually requires a human to organize the data and tell the computer what to look for
Deep Learning is a deeper, multi-layered system that can independently figure out patterns from raw, messy data

Supervised Learning
Like a student learning with a teacher
The model is trained using labeled data meaning every piece of information comes with the correct answer attached
so it can accurately predict the labels for new, unseen data
Examples :
Classifying images (like telling a cat from a dog), detecting email spam, or predicting house prices

Unsupervised Learning
Like a student independently exploring a library to find patterns without any guidance
The model is given raw unlabeled data and is forced to figure things out on its own without right or wrong answers provided
To discover hidden structures, underlying relationships, or to group similar items together (a process known as clustering
Examples: Customer segmentation for marketing, grouping similar news articles, and detecting anomalies in cyber security

Reinforcement Learning
Like a mouse learning to navigate a maze through trial and error
An "agent" interacts with its environment, receiving rewards for good actions and penalties for bad ones
To figure out which actions lead to the best outcomes in order to maximize its cumulative reward over time
Examples: Self-driving cars, robots learning complex movements, and AI mastering complex games like chess or Go

Purpose of Splitting Data
The primary goal is to build robust machine learning models that provide consistent and correct results
A model's performance relies heavily on its training data, so evaluating it properly requires distinct datasets

Standard Split Ratio
A commonly accepted breakdown is 70% for training, 20% for testing, and 10% for validation.

The Three Datasets
Training Data:Used to teach the model and update its parameters. The model optimizes itself based on the "world" represented in this data

Validation Data:Used to evaluate the trained model and fine tune its hyperparameters. This process is repeated until the model shows optimal performance on the validation set
Test Data: A completely unseen dataset used as the final step. It evaluates how well the optimized model generalizes to real-world scenarios

How to Split the Data
Random Split: Used for most standard datasets
Sequential Split:Required for special datasets, like time-based data, where a current data point might depend on a past data point (today's value depending on yesterday's)

Why Use Variable Encodings

Machine learning models require all data to be in numerical format

Encodings are simply the methods we use to convert non-numerical concepts like "cat," "dog" into numbers so the model can process them

Continuous Variables

Definition: Measurable data that naturally falls on a number line, such as weight in kilograms

How to encode: No encoding is necessary because they are already numerical

Nominal Categorical Variables

Definition:Discrete classes that do not have a natural order or ranking, such as "cat" ,"dog" or "zebra"

One-Hot Encoding: Creates a separate question or column for each class. It places a 1 in the correct class column and a 0 in the others

Dummy Encoding:Extremely similar to one-hot encoding, but it leaves out one column

The missing column is unnecessary because if a row has 0 for all other options, it is explicitly implied that the missing column's value must be the correct one

Ordinal Categorical Variables

Definition:Discrete classes that have an obvious, natural order, such as "small," "medium," and "large" coffee sizes

How to encode:While you could technically use one-hot encoding, it treats each class as distinct and loses the natural ordering . Instead, you should map the smallest class to 0, the next to 1, and so on (small = 0, medium = 1, large = 2) to preserve the order

Binary Categorical Variables

Definition:A special case of categorical variables that only have two possible classes, like on/off, fraud/not fraud, or cat/dog

How to encode:Designate one class as the "positive" class and assign it a 1, and make the other class the "negative" class and assign it a 0

Data leakage happens when a machine learning model accidentally learns information from the test data during its training phase
As a result, the model appears to have amazing accuracy during testing, but fails completely when deployed with new data in the real world

It usually occurs when you process your entire dataset before dividing it into training and testing sets . Common mistakes include:
Fixing missing values or scaling features on the whole dataset at once
Randomly splitting time series data, which breaks the chronological order and allows future data to influence past data

How to prevent it?
Split First: Always split your dataset into training and testing sets first, and then process or alter the data for each set separately
Use Cross-Validation:Utilizing techniques like k-fold cross-validation can help prevent the model from overfitting and leaking data

While some people intentionally cause data leakage to get top scores in Kaggle competitions or hackathons, it is considered a bad practice in the IT industry because the models will not work in real production environments